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Original research

The effect of mental fatigue on the performance of Australian football specific skills amongst amateur athletes

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ABSTRACT

Objectives: This study investigated the effects of induced mental fatigue on the performance of Australian football (AF) specific skills amongst amateur AF players.
Design: Randomised cross over trial.
Methods: Twenty-five amateur AF players performed a series of standardised tests from the Australian Football League (AFL) Draft Combine after completing a 30-min Stroop test (mental fatigue condition) or 30-min control condition. The AFL Draft Combine tests included the standing vertical jump test, running vertical jump test, agility test, 20 m sprint, Matthew Lloyd clean hands test, Brad Johnson goal kicking test and a Yo-Yo Intermittent Recovery Level 1 (Yo-Yo IR1) test.
Results: The Stroop test score decreased during the Stroop test (first five trials: mean = 84.7, SD = 3.5; last five trials: mean = 82.2, SD = 5.0, $p = 0.03$). The Yo-Yo IR1 test (mental fatigue: median = 920 m, IQR = 400; control: median = 1040 m, IQR = 760; $p = 0.03$) and Brad Johnson goalkicking test (mental fatigue: median = 19.0, IQR = 5.0; control: median = 25.0, IQR = 10.0, $p = 0.048$) were negatively affected by mental fatigue. No other Draft Combine tests demonstrated a negative affect from mental fatigue.
Conclusions: Mental fatigue had a detrimental influence on the performance of AF specific skills. The findings may have implications for AF players who are required to sustain attention and concentration for prolonged periods before and during matches.

Practical implications

- Mental fatigue leads to impairments in Australian football specific skills (aerobic endurance and kicking accuracy).
- Mental fatigue is likely to contribute to decreases in player performance during Australian football matches.
- Coaches should try to implement strategies to minimise mental fatigue for Australian football players before and during matches.

1. Introduction

Australian football (AF) is a physically demanding intermittent contact sport that involves repeated short bouts of high intensity striding, jumping and sprinting efforts, interspersed with prolonged low-intensity activity periods of walking and jogging.[1] It is a sport that requires well developed physical and technical abilities, but also sustained cognitive attention to perform complex movement patterns, execute tactical plays and make quick accurate decisions in a fast-paced dynamic environment.[2] These cognitive demands suggest AF may cause mental fatigue.

Mental fatigue is a psychobiological state induced by prolonged periods of demanding cognitive activity[3,4] and is associated with low energy levels, difficulties concentrating and lack of motivation.[4,5] Mental fatigue has been found to impair goal-directed attentional capacity,[3] monitoring and adjustment of performance,[6] reaction times, cognitive ability[4] and response to cues in the visual search field for action preparation.[5]

There is emerging evidence to suggest mental fatigue can affect more complex forms of exercise and skill-based tasks. In soccer, a sport comparable to AF in its intermittent activity profile, mental fatigue has been found to impair accuracy and speed of soccer-specific decision-making[7] and impair physical and technical performance in soccer-specific running, passing, tackling and shooting skills.[8,9] The negative effects of mental fatigue on continuous[4] and intermittent running performance[10] have been attributed to higher perceptions of effort.[11] Although there is

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growing evidence to the contrary, findings have suggested mental fatigue also may affect short duration, high-intensity efforts.[12]

The effect of mental fatigue on sports such as soccer, cricket and basketball have previously been investigated[8,12]; however, to date no research has examined the effects of mental fatigue on AF-specific skills. Success in AF demands athletes perform a unique array of physical and technical skills over a time frame of up to 125 min (including breaks).[13] This is important as players must sustain attentional focus for prolonged periods. It is also unclear whether the breaks (6 min between quarters and 20 min for half-time) are long enough to allow recovery from a mentally fatigued state, particularly given the demands of the game and the sustained concentration required during the breaks. In soccer, a defender can impose only so much physical pressure without being called for a foul, however in AF, "tagging", which involves shadowing an opponent all over the field and subjecting him/her to constant physical and verbal harassment is a common strategy within the rules, and is likely to induce added mental fatigue. Given the unique skills and high cognitive demands, the effects of mental fatigue on AF-relevant physical tasks warrants specific investigation.

The Australian Football League (AFL) Draft Combine is an annual gathering, where draftees undergo a battery of standardised tests to assess their prospective recruitment to AFL clubs. The national authoritative body, the AFL, endorses it as a good representation of AF-relevant athletic prowess and technical skills.[14] The purpose of this investigation was to examine the effects of induced mental fatigue on AF specific outcomes. It was hypothesised that mental fatigue would impair players endurance capacity, agility, sprinting ability and AF-specific technical skills.

2. Material and methods

Twenty-five healthy male participants (mean age = 23.8 years, SD = 4.6 years) volunteered to participate in the study. All participants were community-level AF players who were competing in metropolitan or regional Victorian AF leagues in mid/low level divisions. Participants were excluded from the study if they were under 18 or over 25 years of age, were not currently playing AF, had any current injury that would affect the testing, or had surgery within the last 12 months. Participants completed Exercise Sports Science Australia pre-screening assessments (Exercise & Sports Science Australia, 2011) and had to be low risk and safe to participate in vigorous to high intensity exercise.

Based on recent comparable studies on soccer skill performance,[9] a moderate effect size of 0.6 was deemed appropriate and a power analysis indicated that a sample size of 24 would be sufficient to identify within-participant differences (effect size = 0.6, α = 0.05, statistical power = 0.8) using a 2-tailed *t*-test (G*Power, Version 3.1, University of Dusseldorf, Dusseldorf DE).[15] A total of 25 participants were recruited to account for potential dropouts. Participants signed an informed consent form prior to commencing, which outlined the potential risks and study procedures. The study procedures were approved by College of Science, Health and Engineering, La Trobe University Ethics Committee (S17-086).

The study design was a randomised cross over trial, where each participant undertook the experimental and control condition. Participants were tested on three separate occasions, each a minimum of a week apart. The first session was a familiarisation session and the second and third sessions were data collection sessions.

Participants were asked to avoid; vigorous physical activity for 48 h prior to the testing sessions, stressful cognitive tasks, alcohol and caffeine consumption for 3-hs prior to sessions.

During the familiarisation session, participants were given verbal instructions and demonstrations on how to perform each test, including partial completion of the mentally fatiguing task. Borg's 6–20 rating of perceived exertion (RPE) scale was used to measure the participant's physical activity intensity level.[16] During the second session, participants completed one of two conditions (mental fatigue or control), with the condition to be conducted chosen randomly and counterbalanced once half the participants were done. During the third session, participants completed the alternate condition.

For the mentally fatiguing condition, participants completed the 60 s Stroop test 30 times in a row, with approximately 10 s separating each test, for a total of 35 min. The 30 × 60 s Stroop test is recognised and validated as an effective method for inducing mental fatigue.[12] The version of the Stroop test used was a publicly available app-based program called "Brainiversity" (Version 2.0 for personal computer). The Stroop test required participants to answer multiple choice questions by selecting the colour of a word when it was different to the name of the word (e.g., selecting "blue" when the word "red" is printed in blue). The number of correct answers after each 60 s Stroop test was recorded as the participants score for each of the 30 trials. For the control condition participants watched 30 min of a nature documentary that consisted of emotionally neutral content, in a quiet environment.

After completion of each condition and before the AFL draft combine tests began, participants were asked to indicate their subjective rating of motivation using a 5-point Likert scale (0 being "none at all" and 5 being "maximal") for the upcoming performance tests.

Participants performed a 5-min warm up (light run and stretching) before the AFL Draft Combine tests. The AFL Draft Combine performance tests took approximately 50 min to complete and there was a 2-min recovery period between tests. All tests were completed in an indoor sports centre on hardwood floors with participants wearing indoor sports shoes.

Participants completed the standing vertical jump (SVJ) and the running vertical jump (RVJ) tests[14] with a Vertec (JumpUSA™) vertical jump apparatus. Participants completed two trials each of SVJ and RVJ, with 30 s rest between trials. The highest score of the two trials was recorded.

The AFL agility test[14] required participants weave in and out of a course of 5 poles as quickly as possible. Performance times were quantified using dual-beam photoelectric timing gates (Smartspeed timing gate system–Fusion Sport™) and there was a 2-min recovery time between two trials. The fastest of the two trials was recorded.

The 20 m-sprint test[14] required participants to run in a straight line for 20 m as quickly as possible through dual-beam photoelectric timing gates placed at the start and finish of the track (Smartspeed timing gate system–Fusion Sport™). Two trials were completed and the fastest time was recorded.

The Matthew Lloyd clean hands test[14] was used to measure the ability to gather the rolling football cleanly from the ground and deliver quick accurate handballs. A modified version of the standard test was used where participants completed four handballs (with preferred hand) to "target players" at designated positions. The test was reduced from six handballs to four handballs by removing the 10 m target, to shorten test the duration and number of "target players" required. Assessors judged each take and handball on 0–5 scale (0 = fail and 5 = excellent) by following a scoring rubric.[14] The assessor was knowledgeable in AF play and experienced in the use of the scoring rubric.

The Brad Johnson goal kicking test[17] is used to measure AF goal kicking accuracy under different conditions. The test was conducted on a hard wood floor, rather than an outdoor grass oval, to reduce test delays due to unpredictable weather. The indoor sports centre celling height was sufficient to allow normal ball trajectories. The participant took five kicks at goal within 50 s; two drop

Table 1
AFL Draft Combine test results.

Test	Mean + SD		Median [IQR]		p-value	Effect size
	Mental fatigue	Control	Mental fatigue	Control		
Standing vertical jump	61.48 + 5.39	61.24 + 5.08			0.80	0.05
Running vertical jump	73.20 + 7.62	73.72 + 6.19			0.63	0.01
AFL agility (s)	8.46 + 0.40	8.51 + 0.32	8.48 [0.54]	8.60 [0.55]	0.51	−0.13
20 m sprint (s)	3.23 + 0.15	3.22 + 0.30	3.22 [0.20]	3.23 [0.26]	0.38	−0.05
Matthew Lloyd clean hands test (points)	14.72 + 3.69	14.36 + 3.83			0.66	0.10
Brad Johnson kicking test (points)	18.32 + 5.27	20.96 + 6.35	19.00 [5.00]	25.00 [10.00]	0.048	−0.40
Yo-Yo IR1 (m)	1040.00 + 492.75	1182.40 + 537.78	920.00 [400.00]	1040.00 [760.00]	0.03	−0.45
Yo-Yo IR1 HR	183.57 + 19.10	180.35 + 19.40	186.00 [14.00]	187.00 [9.00]	0.40	−0.17
Yo-Yo IR1 RPE	18.17 + 1.05	17.30 + 1.64	18.00 [2.00]	18.00 [3.0]	0.06	−0.38

Note. SD = standard deviation, s = seconds, m = metres, HR = heart rate, RPE = rate of perceived exertion, IQR = interquartile range.

punt kicks from 30 m out on a 45-degree angle, then two snap kicks from 20 m directly in front (one on each foot), and finally a running drop punt kick directly in front from 40 m out. Scoring was based on the AF scoring system (6 points for goal, 1 point for behind, and 0 points for wide of scoring posts). A goal was only awarded if the ball travelled between the two middle posts and above a 2.5 m height marker, otherwise 0 points were awarded. The maximum score was 30 points.

The Yo-Yo IR1 was used to evaluate the participants aerobic endurance.[18] The Yo-Yo IR1 is validated for assessing aerobic endurance and tolerance to incremental and intermittent running exercise and is suggested to be significantly correlated with match running performance.[18] The total distance covered was recorded. Immediately after Yo-Yo IR1 test termination, RPE[16] and heart rate (Polar^{TM} heart rate monitor, Finland) were measured.

Data were analysed using IBM SPSS (Version 26.0, IMB, NY). For normally distributed data, differences between dependent variables were analysed using a paired *t*-test and effect size were calculated by the formula: mean difference divided by the pooled standard deviation ($d = MD/SD_{pooled}$).[19] For non-normally distributed data, differences between dependent variables were analysed using Wilcoxon signed ranks test and effect size were calculated by using the formula: z-score divided by square root of the sample size ($r = z/\sqrt{n}$).[19] Significance was set at $p < 0.05$ (two-tailed) for all analyses.

3. Results

All twenty-five participants who were eligible for the study competed all the tests; there were no dropouts. The average 60 s trial Stroop test score over the first five trials was higher than the average 60 s trial Stroop test score for last five trials (first five trials mean = 84.7, SD = 3.5; last five trials mean = 82.2, SD = 5.0, $p = 0.03$). There was no difference in the level of motivation after the Stroop test and the control condition (Stroop median = 4.0, IQR = 1.0; control = 4.0, IQR = 1.0, $p = 0.59$).

The results for all AFL Combine Draft tests are presented in Table 1. Participants ran a significantly longer distance in the Yo-Yo IR1 test in the control condition than in the mentally fatigued condition. Participants scored significantly more points for the Brad Johnson goalkicking test in the control condition than in the mentally fatigued condition. No other test showed a significant difference between the control and mentally fatigued condition.

4. Discussion

This study investigated the hypothesis that mental fatigue impairs the physical performance of AF-specific skills. The primary findings to support this hypothesis were mental fatigue acutely reduced the performance of the Yo-Yo IR1 test and the Brad Johnson goalkicking test. Mental fatigue had no observable effect on the performance of the other AFL Draft Combine tests.

The decrease in Stroop test scores across the last five trials compared to the first five trials suggests the 30-min based app was likely to be an effective method of inducing fatigue. The level of motivation to perform the AFL Draft Combine tests immediately after the Stroop test (moderately to highly motivated) was similar to the control condition, which contrasts with the theory that motivation is inversely related to mental fatigue.[5] The current finding suggests any changes in performance should be attributed to factors other than motivation differences.

In the present study, participants performed significantly worse on the Yo-Yo IR1 when mentally fatigued. This finding is consistent with previous studies on moderately trained soccer players and elite cricket players that found mental fatigue had a negative impact on the distance covered during the Yo-Yo IR1 test.[8,12] A possible explanation is mental fatigue alters physiological, cardiorespiratory and neuromuscular responses to subsequent exercise.[20] This theory was not supported by the current findings as heart rate was not different between conditions after the Yo-Yo IR1. This aligns with several studies that suggest the adverse effects of mental fatigue are not mediated by changes to peripheral physiological factors or changes to cortico-muscular coupling.[4,10]

A common explanation for why mental fatigue reduces endurance performance is that for the same workload, the perception of effort is higher during the mental fatigue condition. According to psychobiological model of exercise performance, athletes disengage from physical exercise earlier when they perceive effort to be very high or maximal, thus cumulating in shorter total distance covered or time to exhaustion.[21] In this study, RPE at exhaustion was similar after mental fatigue and control conditions, but when mentally fatigued the participants covered less distance to achieve a similar level of RPE, therefore their perception of effort was higher. This finding agrees with the psychobiological model and is consistent the findings of previous studies.[4,8,12]

Participants performed significantly worse on the Brad Johnson goalkicking test when mentally fatigued. The test involved kicking, but also required participants run at submaximal speeds (accelerate/decelerate) and handle the ball. Previous investigations have found mental fatigue impairs aspects of technical performance in soccer-specific simulation protocols of running, decision making, passing accuracy and shooting accuracy.[7] Similarly, reductions in cricket-technical skills performance have been observed with mental fatigue,[12] and mental fatigue has been reported to reduce upper limb stroke performance during table tennis.[22] Given the Brad Johnson goalkicking test involves upper limb (e.g., catching, gripping and release) and lower limb exercise (e.g., running and kicking technique), it is possible mental fatigue could impair the execution of AF-specific technical skills in both upper and lower limbs. It should be noted that while there has been extensive use of the Brad Johnson goalkicking test at the AFL Draft Combine, there has been minimal research conducted on the use of this AF technical skill assessment.[23] Nonetheless, this study found that simulated AF goalkicking is affected by a short 30-min cognitively challenging task.

The current investigation found mental fatigue had no effect on the Matthew Lloyd clean hands test performance; a measure of upper limb technical skill execution (i.e. ball gathering and handballing) and decision-making skills (i.e. react quickly to verbal cues). A possible explanation may relate to the fewer gross motor and physical demands of the clean hands test compared to the goalkicking test, hindering the manifestation of mental fatigue on physical performance. This finding is inconsistent with previous investigations on other upper limb and decision-making skills, which have found mental fatigue impaired table tennis upper limb stroke performance[22] and soccer passing decision-making[7] and reaction times.[3] The AFL's handball test has demonstrated acceptable levels of inter-rater reliability; however, it was suggested further research is needed to confirm the use of the handball test for providing a valid means of handball skill assessment.[24] Therefore, its possible limitations of the test may have contributed to the present finding. Modifying the standard test in this investigation to exclude the 10 m target (to reduce test duration and number of "target players" required) could have further played a factor. The short (6 m) and medium (8 m) distances may have been too simple to elicit meaningful accuracy changes.

This investigation found mental fatigue had no effect on the performance of the SVJ, RVJ, agility test and 20 m-sprint test. These tests are similar in that they are anaerobic rapidly fatiguing tests and performed over a short duration. It has been suggested mental fatigue could reduce voluntary maximal muscle activation level and thus impair force production capacity.[25] In contrast, the current finding is consistent with other studies that showed mental fatigue did not affect maximal strength, power, anaerobic work and voluntary activation levels.[26] Mental fatigue may have a limited effect on the performance of short duration, maximal strength anaerobic tasks because the cognitive component required to complete them is minimal to none.[26] Based on these findings, it appears mental fatigue does not reduce an athlete's ability to maximally use their working muscles for short-duration anaerobic whole-body AF exercise tests.

The limitations of the study should be acknowledged. First, though the technical and physiological tests of the AFL Draft Combine provide valuable information about player skill strengths and weaknesses, they are predominantly performed in a relatively closed setting, separating the skill from the demands of real-world game performance.[23] Second, the Brad Johnson goal kicking test and Matthew Lloyd clean hands test were performed indoor on a hardwood floored surface, wearing sports shoes to avoid delays due to inclement weather, rather than on a grass oval with participants wearing football boots. Further investigations during AF matches should be conducted to verify the relevance of the present findings. Third, only amateur players were tested, so generalisations to elite players should be made with caution. Indeed, the chronic exposure to mental stressors related to professional sport practice, might make elite players more resistant to mental fatigue.[27] Fourth, although performance decrement during a cognitive task (i.e. Stroop test score) is a well-recognised measure of mental fatigue,[28] it is possible that high levels of motivation may allow maintenance of performance under fatiguing conditions, and subjective mental fatigue may precede any decrements in performance.[29] Future investigations should include subjective measures. Lastly, though mental fatigue was induced by a sustained Stroop test, a reliable method of inducing mental fatigue,[12] it does not reflect the real-world ecological environment of an AF game.[30] Future studies should embed mental fatiguing protocols based on more AF specific match activities.

5. Conclusion

The results of this study suggest mental fatigue, induced with a 30-min app-based Stroop test, negatively affected the performance of technical and physical AF-specific skills among amateur athletes. Specifically, mental fatigue acutely impaired aerobic endurance and kicking accuracy, but did not affect shorter and maximal effort AF-specific tasks.

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